

# ELECTRONICS STARTER KIT MANUAL

LEARN • BUILD • CREATE

EXPLORE THE WORLD OF  
ELECTRONICS AND ROBOTICS  
THROUGH HANDS-ON LEARNING AND  
CREATIVE EXPERIMENTATION.



# Starter Kit

## What is a Robotics Starter Kit ?

A robotics starter kit is a beginner-friendly set of tools and components that helps students learn how to build simple robots and electronic projects.

Inside the kit, you will usually find:

- **Electronic parts** (LEDs, resistors, sensors)
- **Movement parts** (motors, wheels)
- **Connecting items** (wires, breadboard)
- **Power source** (battery)

## Why is a Robotics Starter Kit Used?

### Safety Instructions

- Always check connections before powering the circuit
- Do not short-circuit batteries
- Use correct polarity for LEDs and batteries
- Avoid touching exposed wires when powered
- Use tools carefully (wire stripper, screwdriver)
- Do not exceed component voltage ratings

## Kit Components Overview

**LED (Light Emitting Diode):** An LED is a small light that glows when electricity passes through it. It has two legs: a longer positive leg and a shorter negative leg. It is used in indicators, decorations, and simple circuits.

**RGB LED :** An RGB LED combines Red, Green, and Blue LEDs in one package to produce multiple colors. By varying the intensity of each color, you can generate millions of color combinations.

**Buzzer :** A buzzer is a component that produces sound when powered. It is commonly used in alarms and alert systems. It works easily with a battery in simple circuits.

**Vibrator Motor :** A vibrator motor creates vibrations instead of smooth rotation. It is used in mobile phones and small toys. It works when electricity is supplied.

**Transistor :** A transistor is used to control or switch electrical current. It has three legs and can act like a small switch. It is important in many electronic circuits.

**Diode :** A diode allows current to flow in only one direction. It helps protect circuits from wrong connections. It has a marked side to show the negative terminal.

**Capacitor :** A capacitor stores and releases small amounts of electrical energy. It charges and discharges in a circuit. It is used to smooth power supply and signals.

**Potentiometer (Variable Resistor):** A potentiometer is a resistor that can change its value by turning a knob. It is used to control volume, brightness, or speed. It has three terminals.

**Conductive Copper Tape :** Copper tape is used as a flat wire in paper circuits. It conducts electricity and sticks to surfaces easily. It is useful for creative electronics projects.

**Push Button :** A push button is a switch that works only when pressed. It completes the circuit temporarily. It is used in bells, toys, and simple devices.

**Slide Switch :** A slide switch turns a circuit ON or OFF by sliding it. It stays in position after switching. It is commonly used in small electronic devices.

**Toggle Switch** : A toggle switch uses a lever to turn power ON or OFF. It stays in its position until changed. It is used in many electrical systems.

**Rocker Switch** : A rocker switch works by pressing one side to turn ON and the other to turn OFF. It is commonly used in appliances. It is easy to operate.

**Battery** : A battery provides electrical energy to a circuit. It has positive and negative terminals. It is required to power electronic devices.

**Battery Clip** : A battery clip connects the battery to the circuit safely. It holds the battery firmly in place. It makes wiring easier.

**Jumper Wire** : Jumper wires are used to connect components in a circuit. They are easy to use with breadboards. They help in quick testing and building.

**Coin Battery** : A coin battery is a small, flat battery. It provides low voltage, usually 3V. It is used in watches and paper circuits.

**Resistor** : A resistor controls the amount of current flowing in a circuit. It protects components like LEDs from too much current. Its value is shown using color bands.

| COLOR  | 1 <sup>ST</sup> BAND | 2 <sup>ND</sup> BAND | 3 <sup>RD</sup> BAND | MULTIPLIER | TOLERANCE   |
|--------|----------------------|----------------------|----------------------|------------|-------------|
| Black  | 0                    | 0                    | 0                    | 1Ω         |             |
| Brown  | 1                    | 1                    | 1                    | 10Ω        | ± 1% (F)    |
| Red    | 2                    | 2                    | 2                    | 100Ω       | ± 2% (G)    |
| Orange | 3                    | 3                    | 3                    | 1KΩ        |             |
| Yellow | 4                    | 4                    | 4                    | 10KΩ       |             |
| Green  | 5                    | 5                    | 5                    | 100KΩ      | ± 0.5% (D)  |
| Blue   | 6                    | 6                    | 6                    | 1MΩ        | ± 0.25% (C) |
| Violet | 7                    | 7                    | 7                    | 10MΩ       | ± 0.10% (B) |
| Grey   | 8                    | 8                    | 8                    | 100MΩ      | ± 0.05%     |
| White  | 9                    | 9                    | 9                    | 1GΩ        |             |
| Gold   |                      |                      |                      | 0.1Ω       | ± 5% (J)    |
| Silver |                      |                      |                      | 0.01Ω      | ± 10% (K)   |

#### Band order for 4 band:

1st Band → 1st digit 2nd Band → 2nd digit  
3rd Band → Multiplier 4th Band → Tolerance

Example: Red – Violet – Yellow – Gold

- Red = 2
- Violet = 7
- Yellow = ×10,000
- Gold = ±5%

Value =  $27 \times 10,000 = 270,000 \Omega = 270 \text{ k}\Omega \pm 5\%$

#### Band order for 5 band:

1st, 2nd, 3rd = digits

4th = multiplier

5th = tolerance

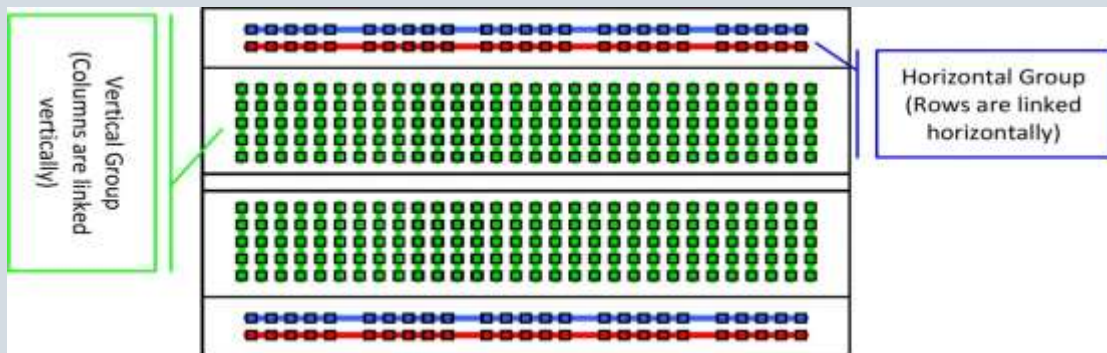
Example: Brown – Black – Red – Orange – Brown

- Brown = 1
- Black = 0
- Red = 2
- Orange = ×1,000
- Brown = ±1%

Value =  $102 \times 1,000 = 102,000 \Omega = 102 \text{ k}\Omega \pm 1\%$

## Breadboard:

A breadboard is used to build circuits without soldering. Components can be inserted and removed easily. It is useful for learning and experimenting. A breadboard is used to build circuits without soldering. Inside it, metal strips connect certain holes together.



## Projects

### 1.Led Glow

#### Components

- ❖ LED
- ❖ Resistor
- ❖ Battery
- ❖ Wires
- ❖ Bread Board

#### Connections

##### 1.Battery

- >9V Battery (+) → Connected to positive rail (red line) on breadboard
- >9V Battery (-) → Connected to negative rail (ground)

## 2.LED

- Anode (+ longer leg) → Connected to positive supply (through wire)
- Cathode (– shorter leg) → Connected to resistor

## 3.Resistor

- One end → Connected to LED cathode
- Other end → Connected to ground (– rail)

## Current Path

Battery (+) → LED → Resistor → Battery (–)

## Working

### 1.Power Supply

- When the battery is connected, voltage is applied across the circuit.

### 2.LED Operation

- Current flows from positive → LED → resistor → ground
- The LED gets forward biased and starts glowing

### 3.Role of Resistor

- Limits current flowing through LED
- Prevents LED from burning out

## Reference Images

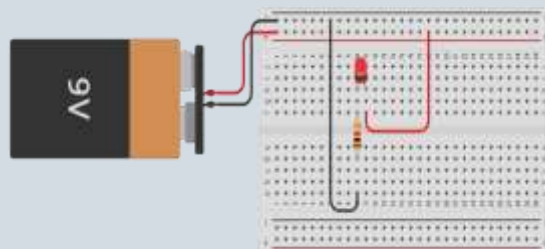


Fig : LED With Resistor

## 2. Led Glow with Paper Circuit

### Components

- LED
- Resistor
- Battery

### Connections

#### 1.Battery (Coin Cell)

- Place the coin cell on the copper tape
- Top side (+) → Connected to top copper tape line
- Bottom side (−) → Connected to bottom copper tape line

#### 2.Copper Tape

- Two strips act like wire:
  - Top strip → Positive (+)
  - Bottom strip → Negative (−)

#### 3.LED Connection

- Long leg (Anode +) → Connected to top copper tape
- Short leg (Cathode −) → Connected to bottom copper tape

### Complete Path

Battery (+) → Copper Tape → LED → Copper Tape → Battery (−)

### Working

#### Step 1: Power Supply

- The coin cell provides voltage to the copper tape.

#### Step 2: Circuit Completion

- When battery touches both copper strips:
  - Circuit becomes closed

#### Step 3: LED Glows

- Current flows through LED
- LED turns ON

### Connections

#### 1.Battery (Coin Cell)

- Place the coin cell on the copper tape
- Top side (+) → Connected to top copper tape line
- Bottom side (−) → Connected to bottom copper tape line

#### 2.Copper Tape

- Two strips act like wires:
  - Top strip → Positive (+)
  - Bottom strip → Negative (−)

### 3.LED Connection

- Long leg (Anode +) → Connected to top copper tape
- Short leg (Cathode -) → Connected to bottom copper tape

### Complete Path

Battery (+) → Copper Tape → LED → Copper Tape → Battery (-)

### Working

#### Step 1: Power Supply

- 1.The coin cell provides voltage to the copper tape.

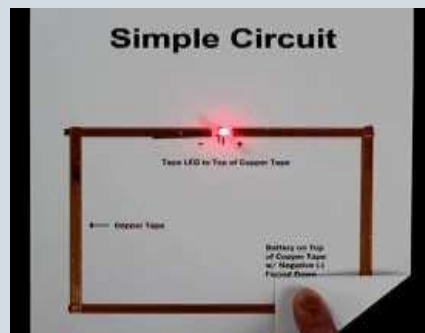
#### Step 2: Circuit Completion

- 2.When battery touches both copper strips:  
→Circuit becomes closed

#### Step 3: LED Glows

- 1.Current flows through LED
- 2.LED turns ON

### Reference Images



### 3.Led Glow with Switch

#### Components

- LED
- Resistor
- Switch
- Battery

#### Connections

##### 1.Battery

- 9V Battery (+) → Connected to push button input
- 9V Battery (-) → Connected to ground rail

##### 2.Push Button

- One side of push button → Connected to battery (+)
- Other side → Connected to LED (through resistor)

### 3.LED

→Anode (+ long leg) → Connected to push button output

→Cathode (– short leg) → Connected to resistor

### 4.Resistor

→One end → Connected to LED cathode

→Other end → Connected to ground (– rail)

### Complete Path

Battery (+) → Push Button → LED → Resistor → Battery (–)

### Working

#### Step 1: Button NOT Pressed

→Circuit is open

→No current flows

→LED is OFF

#### Step 2: Button Pressed

→Push button closes the circuit

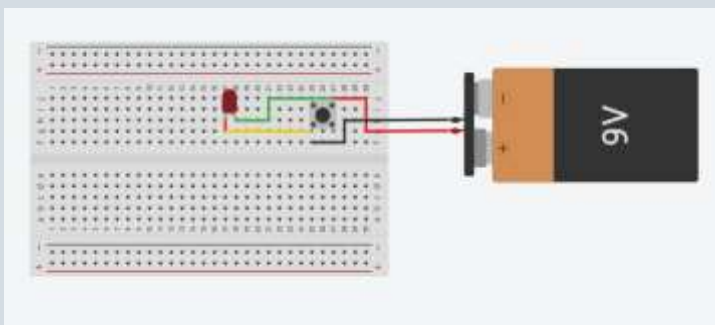
→Current starts flowing

#### Step 3: LED Glows

→Current flows through LED

→LED turns ON only while pressing

### Reference Image



### 3.Alarm with switch

#### Components

- Buzzer
- Switch
- Battery

#### Connections

##### 1.Battery

- 9V Battery (+) → Connected to push button input
- 9V Battery (-) → Connected to ground rail

##### 2.Push Button

- One side → Connected to battery (+)
- Other side → Connected to buzzer (+ terminal)

##### 3.Buzzer

- Positive (+) → Connected to push button output
- Negative (-) → Connected to ground (battery -)

#### Complete Path

Battery (+) → Push Button → Buzzer → Battery (-)

#### Working

##### Step 1: Button NOT Pressed

- Circuit is open
- No current flows
- Buzzer is OFF (no sound)

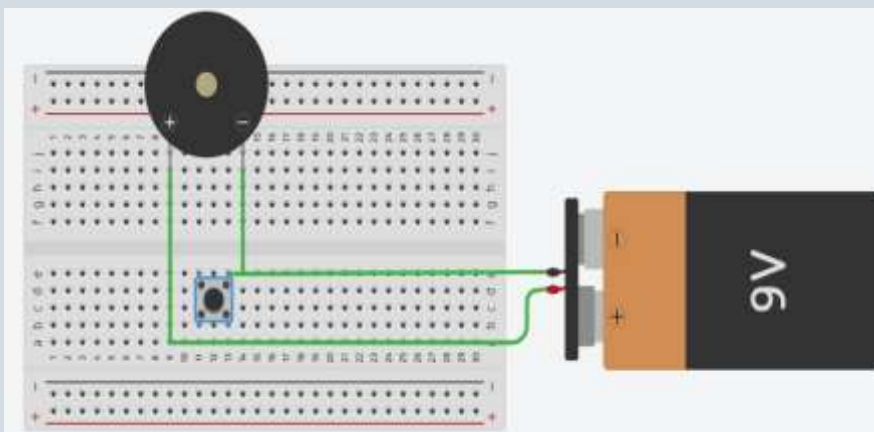
##### Step 2: Button Pressed

- Circuit becomes closed
- Current flows through buzzer

##### Step 3: Sound Produced

- Buzzer converts electrical energy into sound
- You hear a beep sound

#### Reference Image



## 4. Two way switch connection

### Components

- Buzzer
- Switch - 2
- Battery
- Jumper wire

### Connections

#### 1. Battery

- 9V Battery (+) → Connected to positive rail (red line)
- 9V Battery (–) → Connected to ground rail

#### 2. Switches (Slide + Toggle)

- Battery (+) → Connected to first switch (slide switch)
- Output of slide switch → Connected to second switch (toggle switch)
- Output of second switch → Goes to LED

(Switches are connected in series)

#### 3. LED

- Anode (+ long leg) → Connected to output of switches
- Cathode (– short leg) → Connected to resistor

#### 4. Resistor

- One end → Connected to LED cathode
- Other end → Connected to ground (– rail)

### Complete Path

Battery (+) → Switch 1 → Switch 2 → LED → Resistor → Battery (–)

### Working

#### 1. Any Switch OFF

- Circuit is open
- Current cannot flow
- LED remains OFF

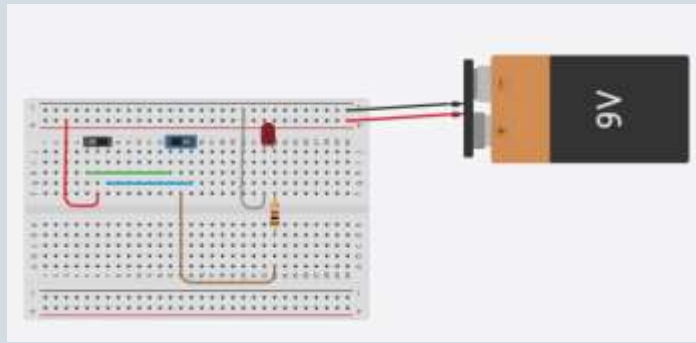
#### 2. Both Switches ON

- Circuit becomes closed
- Current flows through LED

#### 3. LED Turns ON

- LED glows only when both switches are ON

## Reference Image



## 5. Traffic light using LED

### Components

- Resistor
- LED- 3
- Switch
- Battery
- Jumper wire

### Connections

#### 1. Battery

- 9V Battery (+) → Connected to positive rail
- 9V Battery (-) → Connected to ground rail

#### 2. LEDs (3 LEDs)

Each LED is connected in a separate branch:

- Anode (+) → Connected to its respective push button
- Cathode (-) → Connected to resistor → ground

#### 3. Push Buttons (3 switches)

- One side of each button → Connected to positive rail (+)
- Other side → Connected to respective LED

#### 4. Resistors

- Each LED has its own resistor
- Resistor connects LED cathode to ground

### Complete Paths (3 branches)

Battery (+) → Button → LED → Resistor → Battery (-) (Repeated for each LED)

## Working

### 1.No Button Pressed

- All circuits are open
- All LEDs are OFF

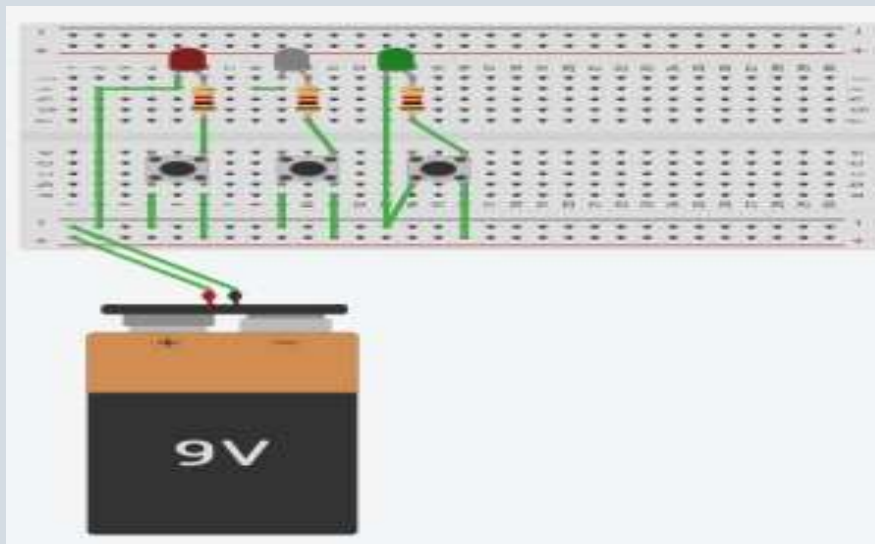
### 2.Press Any One Button

- That branch becomes closed
- Current flows through that LED
- Only that LED glows

### 3.Press Multiple Buttons

- Multiple branches close
- Multiple LEDs glow simultaneously

## Reference Image



## 6.Series Lights using paper circuit

### Components

- LED
- Switch
- Resister
- Copper Tape

## Connections

### 1. Battery (Coin Cell)

- Top side (+) → Connected to top copper tape line
- Bottom side (−) → Connected to bottom copper tape line

### 2. Copper Tape

- Forms a closed rectangular loop
- Acts like wires:
  - Top path → Positive
  - Bottom path → Negative

### 3. LEDs (2 LEDs in Series)

- First LED:
  - Anode (+) → Connected to positive copper tape
  - Cathode (−) → Connected to next LED
- Second LED:
  - Anode (+) → Connected to first LED cathode
  - Cathode (−) → Connected to negative copper tape

## Complete Path

Battery (+) → LED1 → LED2 → Battery (−)

## Working

### 1. Power Supply

- Coin cell provides voltage to the copper tape

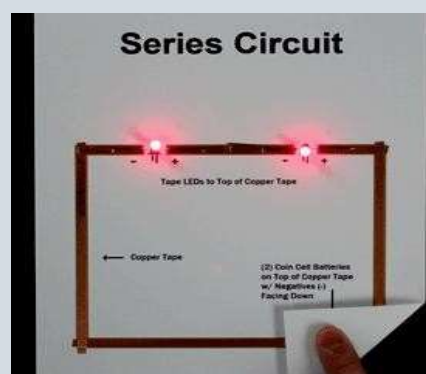
### 2. Series Connection Behavior

- Current flows through both LEDs one after another
- Same current passes through both LEDs

### 3. LEDs Glow

- Both LEDs glow together
- If one LED is removed or fails → circuit breaks → both OFF

## Reference Image



## 7.Parallel Lights using paper circuit

### Components

- LED
- Switch
- Resister
- Copper Tape

### Connections

#### 1.Battery (Coin Cell)

- Top side (+) → Connected to top copper tape line
- Bottom side (-) → Connected to bottom copper tape line

#### 2.Copper Tape Layout

- Two main horizontal lines:
  - Top line → Positive (+)
  - Bottom line → Negative (-)
- Vertical copper paths connect LEDs between these two lines

#### 3.LEDs (Parallel Connection)

Each LED is connected separately:

- Anode (+) → Connected to top copper tape
- Cathode (-) → Connected to bottom copper tape

Each LED has its own path to the battery

### Complete Paths

Battery (+) → LED1 → Battery (-) Battery (+) → LED2 → Battery (-)

(Multiple independent paths)

### Working

#### 1.Power Supply

- Coin cell provides voltage across both copper lines

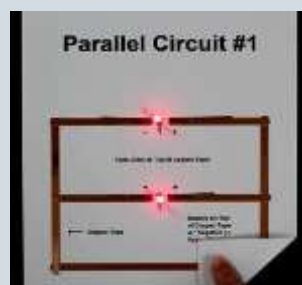
#### 2.Parallel Behavior

- Each LED gets the same voltage
- Current splits into multiple paths

#### 3.LEDs Glow

- All LEDs glow independently
- If one LED fails → others still work

### Reference Image



## 8.Series Lights using LED circuit

### Connections (Series Circuit)

#### 1.Battery Connections

- Battery Positive (+) → connected to first LED anode (+)
- Battery Negative (−) → connected to resistor → last LED cathode (−)

#### 2.LED Series Connections

In a series circuit, components are connected end-to-end:

- LED 1 cathode (−) → connected to LED 2 anode (+)
- LED 2 cathode (−) → connected to LED 3 anode (+)
- Continue this chain...

So current flows through all LEDs one after another

#### 3.Resistor Placement

- A single resistor is used in series
- Connected:
  - Either before first LED OR
  - after last LED (as shown in your image)

It limits current for the entire chain Working of the

Circuit

#### 1.Power Flow Starts

- Battery provides voltage
- Current flows from + → through LEDs → resistor → −

#### 2.LEDs Glow Sequentially

- Current passes through LED 1 → LED 2 → LED 3
- Each LED lights up as current flows through it

#### 3.Series Behavior

- Same current flows through all components
- Voltage gets divided across LEDs

Example:

- 9V battery with 3 LEDs → each LED gets ~3V (approx)

### Reference Image



## 9.Parallel Lights using LED circuit

### Connections

#### 1.Battery Connections

- breadboard
- 9V Battery Positive (+) → Connect to red rail (top power line) on
  - 9V Battery Negative (−) → Connect to blue rail (bottom power line)

#### 2.Resistor Connections

Each LED must have its own resistor.

- One end of each resistor → connected to positive rail (+)
- Other end of resistor → goes to LED anode (+)

This limits current and protects LEDs.

#### 3.LED Connections

Each LED has two legs:

- Long leg (Anode +)
- Short leg (Cathode −)

Connections:

- Anode (long leg) → connected to resistor output
- Cathode (short leg) → connected to ground rail (−)

#### 4.Parallel Setup

All LEDs are connected in parallel, meaning:

- All anodes connect to + via resistors
- All cathodes connect to − (ground rail)

### Working of the Circuit

#### 1.Power Supply Starts

- The 9V battery provides voltage to the breadboard rails
- Current flows from positive → negative

#### 2. Current Through Resistor

- Current first passes through each resistor
- Resistor reduces current to safe levels

#### 3. LED Glows

- Current enters LED anode (+)
- Flows through LED → exits via cathode (−)
- LED emits light

## 4.Parallel Setup

All LEDs are connected in parallel, meaning:

- All anodes connect to + via resistors
- All cathodes connect to – (ground rail)

## Working of the Circuit

### 1. Power Supply Starts

- The 9V battery provides voltage to the breadboard rails
- Current flows from positive → negative

### 2. Current Through Resistor

- Current first passes through each resistor
- Resistor reduces current to safe levels

### 3. LED Glows

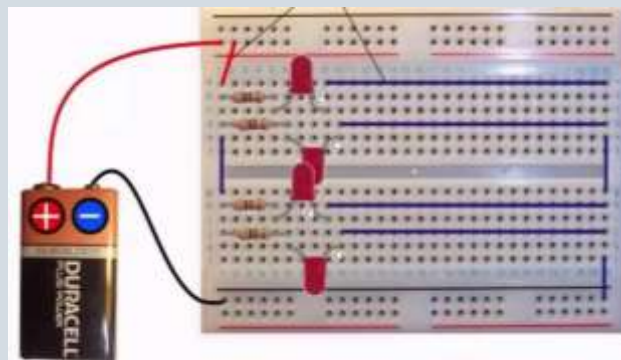
- Current enters LED anode (+)
- Flows through LED → exits via cathode (–)
- LED emits light

### 4.Parallel Behavior

- Each LED gets same voltage (9V)
- If one LED fails → others still glow



## Reference Image



## 10.LED with Capacitor and transistor

### Components

LED  
Transistor  
Resister  
Battery  
Capacitor

### Connections

#### 1.Battery Connections

- 9V Battery (+) → Breadboard positive rail
- 9V Battery (–) → Breadboard ground rail

#### 2.LED Connection

- LED Anode (+ longer leg) → Connected to Vcc (+ rail) through a resistor
- LED Cathode (– shorter leg) → Connected to Collector (C) of BC547

#### 3.Transistor

(Flat side facing you → pins are Collector – Base – Emitter)

- Collector (C) → LED negative side
- Emitter (E) → Ground (– rail)
- Base (B) → Connected to resistor + capacitor junction

#### 4.Capacitor

- One terminal → Connected to Base of transistor
- Other terminal → Connected to Ground

#### 5.Resistor (Base Resistor)

- One end → Vcc (+ rail)
- Other end → Base of transistor

#### 6.Extra Resistor (LED protection)

In series with LED (important to prevent damage)

### Working

#### Step 1: Power ON

- The battery supplies voltage.
- The capacitor starts charging through the resistor.

## Step 2: Delay Creation

- Capacitor voltage increases slowly.
- The transistor remains OFF initially.

## Step 3: Transistor Turns ON

- When capacitor voltage reaches  $\sim 0.7V$ :
  - Base gets enough current
  - Transistor switches ON

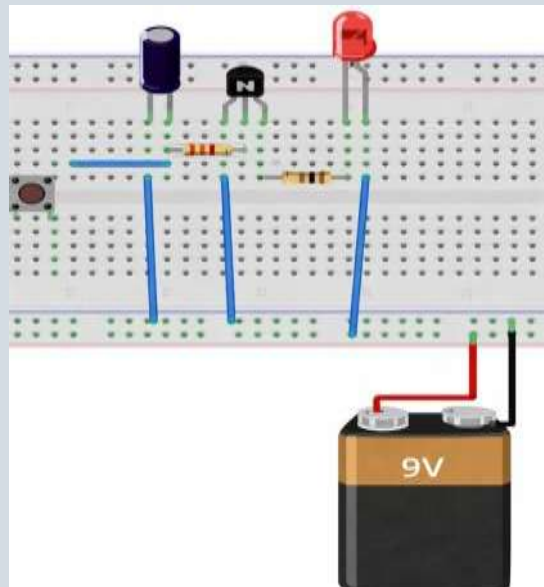
## Step 4: LED Glows

- Current flows:  
 $V_{cc} \rightarrow \text{LED} \rightarrow \text{Transistor} \rightarrow \text{Ground}$
- LED turns ON after delay

## Step 5: Discharge (if power removed)

- Capacitor discharges → transistor OFF → LED OFF

## Reference Image



## 11.Adjustable LED with Potentiometer

### Components

- LED
- Potentiometer
- Battery
- Jumper Wires

### Working

The **battery** provides power to the circuit.

- The **LED** lights up when current flows through it.
- The **potentiometer** controls the amount of current going to the LED.

### Step-by-step working:

#### Connection Setup

- One end of the potentiometer → connected to battery (+)
- Other end → connected to ground (-)
- Middle pin (wiper) → connected to the LED (through a resistor if

needed)

#### Adjusting the Knob

- When you rotate the potentiometer knob, it changes resistance.

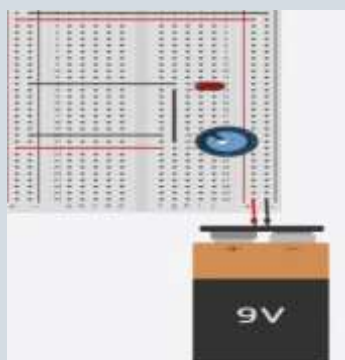
#### Low Resistance

- More current flows → LED glows bright

#### High Resistance

- Less current flows → LED becomes dim

### Reference Image



## 12. Working with a Multimeter?

A multimeter is a tool used to measure:

- Voltage (V)
- Current (A)
- Resistance ( $\Omega$ )
- Continuity (beep test)

### Basic Setup

- Black probe → COM (common)
- Red probe → V $\Omega$ mA port
- Turn the dial to the required mode

### 1. Measuring Voltage (Battery) Steps:

- Set dial to DC Voltage (V $\overline{\text{=}}$ )

#### 1. Place:

- Red probe → Positive (+)
- Black probe → Negative (-)

#### 2. Read value on display

### Working:

- Multimeter measures potential difference
- Example: 9V battery → shows around 9V

### 3. Continuity Test Steps:

1. Set dial to continuity mode
2. Touch probes across wire/component

### Working:

- If path is complete → beep sound
- If broken → no sound

### 4. Measuring Current (Advanced) Steps:

1. Move red probe to A or mA port
2. Break circuit and connect multimeter in series
3. Set dial to A

### Working:

- Measures flow of current through circuit

### Simple Understanding

Voltage = Pressure Current = Flow Resistance = Opposition

### Safety Tips

- Always start with highest range
- Never measure resistance in a powered circuit
- Check probe connections before measuring current

### Package Includes:

1 x Electronics Component Starter Kit

## Kit includes:

20 x LED (Light Emitting Diode) – Multicolor (Red, Green, Blue, White, Yellow, RGB), 3mm  
1 x Buzzer (6V–12V, 2.8 kHz)  
1 x Vibrator Motor (3V DC, 9000 RPM) 5 x BC547  
Transistors (NPN)  
5 x Diodes  
1 x Resistor Kit (30 Values)  
5 x Capacitors (Mixed Values) 1 x  
Potentiometer (10K Ohm)  
1 x Copper Tape (10mm Conductive)  
3 x Push Buttons (Tactile Switch) 2 x Slide  
Switch (SPDT)  
2 x Toggle Switch  
3 x Rocker Switch 1 x  
9V Battery  
1 x Battery Clip (9V Connector)  
60 x Jumper Wires (20 M-M, 20 M-F, 20 F-F)  
1 x Breadboard (400 Points)  
2 x Coin Cell Batteries (CR2025) 1 x Plastic  
Storage Box

